Evolutionary Competition of AI Governance Architectures in the Standard Nuclear oliGARCHy

Soumadeep Ghosh

Kolkata, India

Abstract

This paper develops a rigorous mathematical framework for evaluating artificial intelligence governance architectures embedded within a multi-district institutional system called the *Standard Nuclear oliGARCHy* (SNoG). Governance algorithms are modeled as competing institutional technologies whose adoption evolves through stochastic replicator dynamics coupled with GARCH-type conditional volatility. The framework integrates stochastic optimal control, evolutionary game theory, mean-field institutional adoption, and macroeconomic feedback to produce a tractable yet rich model of governance competition.

We establish four main analytical results. First, we prove a dominance theorem: if a unique architecture maximizes risk-adjusted fitness, its population share converges to unity almost surely. Second, we characterize interior equilibria under switching costs using a mean-field logit formulation and derive conditions for coexistence versus monoculture. Third, we derive a tight bound on the expected time to governance dominance and its dependence on the fitness gap, district count, and outcome variance. Fourth, we prove a social welfare loss theorem quantifying the aggregate cost of architectural fragmentation under a utilitarian social planner. We complement the theory with a calibrated simulation study comparing RLHF, Constitutional AI, and a Hybrid architecture. The simulations demonstrate that Constitutional AI achieves the highest risk-adjusted fitness despite its lower raw mean welfare, because its governance outcomes are exceptionally stable. By contrast, the Hybrid architecture suffers a severe fitness penalty from volatility clustering, illustrating that raw performance is not the decisive determinant of long-run institutional dominance.

The SNoG thus functions as a *governance discovery mechanism*—a laboratory that allows alignment strategies to reveal their institutional properties through observable outcomes rather than theoretical claims alone.

The paper ends with "The End"

Contents

1 Introduction **2**
- 1.1 Motivation . . . 2
- 1.2 The SNoG Framework . . . 3
- 1.3 Contributions . . . 3
- 1.4 Outline . . . 3

2 The Standard Nuclear oliGARCHy **3**
- 2.1 Structural Definition . . . 3
- 2.2 The Nuclear Governance Council . . . 4
- 2.3 Inter-District Spillovers . . . 4

3 District Economic Dynamics **4**
- 3.1 State Space and Transition . . . 4
- 3.2 Governance Policy . . . 4
- 3.3 Inter-District Spillovers in Dynamics . . . 4

4 District Welfare and Risk-Adjusted Fitness **5**
4.1 Instantaneous Welfare . . . 5
4.2 Intertemporal Welfare . . . 5
4.3 Mean Welfare Across Districts . . . 5
4.4 Risk-Adjusted Fitness . . . 5

5 Stochastic Replicator Dynamics **5**
5.1 Deterministic Baseline . . . 5
5.2 Stochastic Extension . . . 6

6 Institutional Volatility: The oliGARCH Process **6**
6.1 Motivation . . . 6
6.2 The oliGARCH(1,1) Specification . . . 6
6.3 Volatility and Fitness Interaction . . . 6

7 Main Theoretical Results **7**
7.1 Dominance and Convergence . . . 7
7.2 Expected Time to Governance Dominance . . . 8
7.3 Social Welfare Loss from Fragmentation . . . 8

8 Mean-Field Equilibria and Coexistence **8**
8.1 Logit Adoption with Switching Costs . . . 8
8.2 Effect of the NGC on Equilibrium . . . 9

9 Simulation Study **9**
9.1 Calibration . . . 9
9.2 Algorithm . . . 10
9.3 Simulation Results . . . 10
9.4 Figures . . . 10

10 Macroeconomic Implications **11**
10.1 Governance and Aggregate Output . . . 11
10.2 Volatility Clustering and Systemic Risk . . . 11

11 Conclusion **11**

List of Figures

1 Representative adoption trajectory on the SNoG governance simplex. . . . 10
2 Stylized conditional variance paths for each governance architecture under the oliGARCH(1,1) specification. . . . 11

List of Tables

1 Calibrated oliGARCH Parameters . . . 9
2 Simulation Results ($T = 200$, $K = 3$, $N_{\text{sim}} = 5{,}000$) . . . 10

1 Introduction

1.1 Motivation

As artificial intelligence systems grow more capable, the question of which governance architecture should oversee AI-assisted decision-making acquires institutional importance beyond the realm of computer science. Competing alignment strategies—reinforcement learning from human feedback [3], constitutional approaches [1], and hybrid designs—differ not only in their formal properties but in the observable economic and political outcomes they generate when deployed at scale.

Classical alignment research evaluates governance architectures primarily through theoretical safety arguments or controlled laboratory benchmarks. These methods have clear merit but face a fundamental

limitation: real-world governance operates under persistent uncertainty, endogenous feedback, and competitive pressure from alternative designs. A framework that embeds governance algorithms within an evolving institutional environment, where performance is adjudicated by aggregate outcomes rather than axioms, offers a complementary and practically grounded perspective.

1.2 The SNoG Framework

We model such an environment as a multi-district polity, the *Standard Nuclear oliGARCHy* (SNoG). The name reflects two structural features. The *nuclear* component refers to a small central authority—the Nuclear Governance Council (NGC)—that sets meta-level constitutional constraints binding on all districts. The *oliGARCH* component refers both to the oligopolistic character of architecture competition (a small number of rival designs compete for institutional adoption) and to the GARCH-type conditional volatility that governs governance outcomes, directly embedding the financial econometrics of Engle [4] into the institutional model.

Districts within the SNoG adopt governance architectures independently. Adoption shares evolve according to stochastic replicator dynamics: architectures that generate above-average welfare gain institutional market share, while underperformers lose it. The SNoG thereby functions as a *decentralized discovery mechanism* for alignment strategies, surfacing governance quality through competition rather than prescription.

1.3 Contributions

This paper makes four contributions.

(i) **Formal SNoG structure.** We provide a precise definition of the SNoG, including district dynamics, inter-district spillovers, and the role of the NGC, building on mean-field game theory [7] and evolutionary game theory [5].

(ii) **Dominance theorem with full proof.** We prove that strict fitness dominance implies almost-sure convergence to architectural monoculture, using Itô's formula and a Lyapunov argument for stochastic differential equations.

(iii) **New equilibrium and welfare results.** We characterize coexistence equilibria under switching costs, derive a convergence rate bound, and prove a social welfare loss theorem for fragmented governance.

(iv) **Calibrated simulation.** We implement and calibrate the model with empirically motivated parameters, reporting adoption trajectories, volatility paths, and welfare comparisons across three real-world governance architectures.

1.4 Outline

Section 2 defines the SNoG formally. Section 3 specifies district economic dynamics and optimal control. Section 4 introduces welfare and fitness. Sections 5–6 develop replicator dynamics and institutional volatility. Section 7 states and proves the main theoretical results. Section 8 derives mean-field equilibria and coexistence conditions. Section 9 presents the calibrated simulation. Section 10 discusses macroeconomic implications. Section 11 concludes.

2 The Standard Nuclear oliGARCHy

2.1 Structural Definition

Definition 1 (SNoG)**.** A *Standard Nuclear oliGARCHy* is a tuple

$$\mathcal{G} = \big(\mathcal{D},\, \mathcal{K},\, \mathcal{C},\, \Phi,\, \mathcal{N}\big),$$

where:

- $\mathcal{D} = \{1, \ldots, D\}$ is the finite set of districts;
- $\mathcal{K} = \{1, \ldots, K\}$ is the finite set of competing governance architectures;

- $\mathcal{C}$ is the constitutional constraint set enforced by the NGC;
- $\Phi : \mathcal{K} \to \mathbb{R}$ is the fitness mapping defined in Section 4.4;
- $\mathcal{N}$ is the inter-district spillover network.

2.2 The Nuclear Governance Council

The NGC occupies the center of the SNoG. It does not choose district-level governance but enforces a set of *constitutional constraints* $\mathcal{C}$—minimum welfare floors, volatility ceilings, and due-process requirements—that any architecture must satisfy to remain eligible for adoption. Formally, let $\mathcal{K}^* \subseteq \mathcal{K}$ denote the set of constitutionally admissible architectures:

$$\mathcal{K}^* = \big\{k \in \mathcal{K} : \Phi(k) \text{ satisfies all constraints in } \mathcal{C}\big\}.$$

All subsequent analysis restricts attention to $\mathcal{K}^*$. The NGC therefore narrows the competitive arena without determining its outcome.

2.3 Inter-District Spillovers

Districts are not isolated. Governance quality spills across district boundaries through policy diffusion, regulatory arbitrage, and demonstration effects. We represent these links by a weighted adjacency matrix $G \in \mathbb{R}^{D\times D}$, where $G_{ij} \geq 0$ captures the spillover intensity from district j to district i, with $G_{ii} = 0$ and rows normalized so $\sum_j G_{ij} \leq 1$. Inter-district spillovers enter district welfare additively (Section 4) and modify the effective fitness landscape (Section 4.4).

3 District Economic Dynamics

3.1 State Space and Transition

Let $x_t^d \in \mathbb{R}^n$ denote the state vector of district d at time t, aggregating economic indicators such as output, price stability, and institutional trust. District dynamics follow the controlled linear system

$$x_{t+1}^d = A\, x_t^d + B\, a_t^d + \Gamma\, z_t^d + \varepsilon_t^d, \tag{1}$$

where $A \in \mathbb{R}^{n\times n}$ is the transition matrix, $B \in \mathbb{R}^{n\times m}$ maps governance policy $a_t^d \in \mathbb{R}^m$ to state dynamics, $z_t^d \in \mathbb{R}^p$ is a vector of exogenous macroeconomic shocks with loading matrix Γ, and $\varepsilon_t^d \sim \mathcal{N}(0, \Sigma_\varepsilon)$ is idiosyncratic district noise.

Assumption 1 (Stability)**.** The pair (A, B) is stabilizable: there exists a matrix L such that $A - BL$ has all eigenvalues strictly inside the unit circle.

3.2 Governance Policy

Architecture $k \in \mathcal{K}^*$ generates policy through a mapping

$$a_t^d = \pi_k\big(x_t^d\big),$$

where $\pi_k : \mathbb{R}^n \to \mathbb{R}^m$ is the *policy function* associated with architecture k. Under the linear-quadratic structure of Section 4, the optimal policy function for architecture k takes the linear feedback form

$$\pi_k(x_t^d) = -L_k\, x_t^d, \tag{2}$$

where $L_k = (R_k + \beta B^\top P_k B)^{-1}\beta B^\top P_k A$ and P_k solves the associated discrete-time algebraic Riccati equation (DARE):

$$P_k = Q_k + \beta A^\top P_k A - \beta^2 A^\top P_k B(R_k + \beta B^\top P_k B)^{-1} B^\top P_k A. \tag{3}$$

3.3 Inter-District Spillovers in Dynamics

Incorporating spillovers, district dynamics become

$$x_{t+1}^d = A\, x_t^d + B\, \pi_k(x_t^d) + \delta \sum_{j\neq d} G_{dj}\, x_t^j + \varepsilon_t^d, \tag{4}$$

where $\delta \geq 0$ is a spillover intensity parameter. When $\delta = 0$ the model reduces to the isolated-district case in (1).

4 District Welfare and Risk-Adjusted Fitness

4.1 Instantaneous Welfare

Instantaneous district welfare under architecture k is represented by the quadratic loss

$$U(x_t, a_t; k) = -x_t^\top Q_k\, x_t - a_t^\top R_k\, a_t, \tag{5}$$

where $Q_k \succeq 0$ and $R_k \succ 0$ are architecture-specific weighting matrices encoding different policy priorities (e.g., stability versus responsiveness).

Assumption 2 (Welfare matrices)**.** For every $k \in \mathcal{K}^*$, Q_k is positive semidefinite and R_k is positive definite, so that (3) has a unique positive-semidefinite solution P_k.

4.2 Intertemporal Welfare

The expected intertemporal welfare of district d under architecture k, starting from initial state x_0^d, is

$$W_k(x_0^d) = \mathbb{E}\left[\sum_{t=0}^{\infty} \beta^t\, U(x_t^d, \pi_k(x_t^d); k) \,\middle|\, x_0^d\right], \tag{6}$$

where $\beta \in (0,1)$ is the common discount factor. Substituting the optimal policy (2) and using the DARE solution P_k:

$$W_k(x_0^d) = -(x_0^d)^\top P_k\, x_0^d - \frac{\beta}{1-\beta}\,\mathrm{tr}(P_k \Sigma_\varepsilon). \tag{7}$$

4.3 Mean Welfare Across Districts

For the purposes of evolutionary competition, we model realized district welfare under architecture k as

$$\widetilde{W}_{k,t} = \mu_k + \varepsilon_{k,t}, \tag{8}$$

where $\mu_k = \mathbb{E}[W_k]$ is the steady-state mean welfare and $\varepsilon_{k,t}$ is a zero-mean shock whose conditional variance $h_{k,t}$ follows the oliGARCH process of Section 6.

4.4 Risk-Adjusted Fitness

Definition 2 (Institutional fitness)**.** The *risk-adjusted institutional fitness* of architecture k is

$$\phi_k = \mu_k - \tfrac{1}{2}\,\sigma_k^2, \tag{9}$$

where $\sigma_k^2 = \mathbb{E}[h_{k,t}]$ is the unconditional variance of governance outcomes.

The fitness measure (9) is the continuous-time analogue of a log-expected-welfare criterion; it penalizes variance because districts are risk-averse with respect to governance instability. This parallels the Kelly criterion and the log-growth-rate decomposition in portfolio theory. As will be seen in the proof of Theorem 1, the expression ϕ_k arises *endogenously* from Itô's formula applied to the log-share ratio: architectures that use raw mean welfare μ_k in the replicator drift are automatically penalized by $\frac{1}{2}\sigma_k^2$ through the quadratic variation of the noise process.

Remark 1. The fitness function (9) encodes the mean-variance trade-off precisely: architectures with high raw performance but also high volatility may be dominated by more stable alternatives. This formalizes the intuition that governance reliability—not merely expected output—determines long-run institutional success.

5 Stochastic Replicator Dynamics

5.1 Deterministic Baseline

Let $s_k(t) \in (0,1)$ denote the share of districts adopting architecture k at time t, with $\sum_k s_k(t) = 1$. The deterministic replicator equation is

$$\dot{s}_k = s_k\big(\mu_k - \bar{\mu}\big), \qquad \bar{\mu}(t) = \sum_{j \in \mathcal{K}^*} s_j(t)\,\mu_j. \tag{10}$$

This is the standard replicator dynamic from evolutionary game theory [5, 8]: architectures outperforming the population mean gain share and those underperforming lose it.

5.2 Stochastic Extension

Incorporating outcome uncertainty, replicator dynamics become the system of Itô stochastic differential equations

$$ds_k = s_k\big(\mu_k - \bar{\mu}\big)dt + s_k\,\sigma_k\,dB_k(t) - s_k\,\bar{\sigma}\,d\bar{B}(t), \tag{11}$$

where $B_k(t)$ are independent standard Brownian motions, $\bar{\mu} = \sum_j s_j\mu_j$ is the population mean welfare, $\bar{\sigma}\,d\bar{B} = \sum_j s_j\sigma_j\,dB_j$ is the mean noise term, and all stochastic integrals are in the Itô sense.

Remark 2. The stochastic term in (11) preserves the simplex Δ^{K-1}: given $\sum_k s_k = 1$ and $\sum_k ds_k = 0$, the population shares remain a valid probability distribution for all $t \geq 0$ almost surely. Note that the drift uses raw mean welfare μ_k, not the risk-adjusted fitness ϕ_k. The variance penalty in ϕ_k arises endogenously from Itô's lemma, as demonstrated in the proof of Theorem 1.

6 Institutional Volatility: The oliGARCH Process

6.1 Motivation

Governance outcomes are not i.i.d. over time. Institutional crises cluster: a period of policy turbulence under architecture k tends to be followed by further turbulence. Conversely, stable regimes exhibit persistence. This volatility clustering is the institutional analogue of the ARCH effects documented in financial returns by Engle [4] and generalized by Bollerslev [2].

6.2 The oliGARCH(1,1) Specification

Let $\varepsilon_{k,t}$ be the welfare innovation for architecture k at time t. Conditional on the filtration $\mathcal{F}_{t-1}$:

$$\varepsilon_{k,t} \mid \mathcal{F}_{t-1} \sim \mathcal{N}(0,\, h_{k,t}), \tag{12}$$

$$h_{k,t} = \omega_k + \alpha_k\,\varepsilon_{k,t-1}^2 + \beta_k\,h_{k,t-1}, \tag{13}$$

with parameters satisfying the following assumption.

Assumption 3 (oliGARCH stationarity)**.** For every $k \in \mathcal{K}^*$: $\omega_k > 0$, $\alpha_k \geq 0$, $\beta_k \geq 0$, and $\alpha_k + \beta_k < 1$.

Under Assumption 3, the unconditional variance exists and equals

$$\sigma_k^2 = \mathbb{E}[h_{k,t}] = \frac{\omega_k}{1 - \alpha_k - \beta_k}. \tag{14}$$

The persistence of volatility shocks is captured by $\alpha_k + \beta_k$: values close to one imply that welfare crises are long-lived, while low values imply rapid mean-reversion to baseline stability.

6.3 Volatility and Fitness Interaction

Under the oliGARCH specification, the risk-adjusted fitness (9) can be written explicitly as

$$\phi_k = \mu_k - \frac{\omega_k}{2(1 - \alpha_k - \beta_k)}. \tag{15}$$

This expression makes the mechanism precise: governance architectures with high baseline welfare μ_k but also high shock variance ω_k or high volatility persistence $\alpha_k + \beta_k$ suffer a fitness penalty. The trade-off between mean performance and stability is central to the evolutionary selection process.

7 Main Theoretical Results

7.1 Dominance and Convergence

Theorem 1 (Governance dominance)**.** *Under Assumptions 1–3, if there exists a unique architecture $k^* \in \mathcal{K}^*$ such that*

$$\phi_{k^*} > \phi_j \quad \text{for all } j \neq k^*,$$

then $s_{k^*}(t) \to 1$ almost surely as $t \to \infty$.

Proof. Fix any competing architecture $j \neq k^*$ and define the log-share ratio

$$Y_t^j = \log \frac{s_{k^*}(t)}{s_j(t)}. \tag{16}$$

We apply Itô's formula to $f(s) = \log s$, which gives $d(\log s_k) = s_k^{-1}\, ds_k - \frac{1}{2} s_k^{-2} (ds_k)^2$. Substituting the stochastic replicator (11) for both s_{k^*} and s_j, and computing the quadratic variation $(s_k\, \sigma_k\, dB_k - s_k\, \bar{\sigma}\, d\bar{B})^2 = s_k^2(\sigma_k^2 + \bar{\sigma}^2)\, dt$, one obtains

$$dY_t^j = \Big[\big(\mu_{k^*} - \tfrac{1}{2}\sigma_{k^*}^2\big) - \big(\mu_j - \tfrac{1}{2}\sigma_j^2\big)\Big] dt + \sigma_{k^*}\, dB_{k^*} - \sigma_j\, dB_j, \tag{17}$$

where the $\bar{\sigma}$ terms cancel in the difference $d(\log s_{k^*}) - d(\log s_j)$. Recognizing the definitions $\phi_{k^*} = \mu_{k^*} - \frac{1}{2}\sigma_{k^*}^2$ and $\phi_j = \mu_j - \frac{1}{2}\sigma_j^2$, equation (17) simplifies to

$$dY_t^j = \Delta\phi_j\, dt + \tilde{\sigma}_j\, d\widetilde{B}_t, \tag{18}$$

where $\Delta\phi_j = \phi_{k^*} - \phi_j > 0$ by hypothesis, $\tilde{\sigma}_j^2 = \sigma_{k^*}^2 + \sigma_j^2$, and $\widetilde{B}_t$ is the standard Brownian motion $(\sigma_{k^*} B_{k^*} - \sigma_j B_j)/\tilde{\sigma}_j$.

Equation (18) describes arithmetic Brownian motion with positive drift $\Delta\phi_j > 0$. By the strong law of large numbers for Brownian motion, $\widetilde{B}_t/t \to 0$ almost surely, so

$$\frac{Y_t^j}{t} \;\to\; \Delta\phi_j > 0 \quad \text{a.s.}$$

Therefore $Y_t^j \to +\infty$ almost surely, which implies $s_j(t)/s_{k^*}(t) \to 0$ almost surely. Since this holds for every $j \neq k^*$ and shares sum to one, $s_{k^*}(t) \to 1$ almost surely. □

Corollary 1 (Speed of convergence)**.** *Under the conditions of Theorem 1, the probability that $s_{k^*}(t) < 1-\delta$ decays exponentially in t for any $\delta > 0$:*

$$\mathbb{P}\big(s_{k^*}(t) < 1 - \delta\big) \leq C \exp\left(-\frac{\Delta\phi_{\min}^2}{2\,\tilde{\sigma}_{\max}^2}\, t\right),$$

where $\Delta\phi_{\min} = \min_{j \neq k^*}(\phi_{k^*} - \phi_j)$, $\tilde{\sigma}_{\max}^2 = \max_j(\sigma_{k^*}^2 + \sigma_j^2)$, and $C > 0$ is a constant depending on δ and the initial conditions.

Proof. From (18), $Y_t^j \sim \mathcal{N}(\Delta\phi_j\, t,\, \tilde{\sigma}_j^2\, t)$ for each $j \neq k^*$. The event $\{s_{k^*}(t) < 1 - \delta\}$ implies that at least one ratio s_j/s_{k^*} exceeds $\delta/(1-\delta)$, i.e., $Y_t^j < \log\big((1-\delta)/\delta\big)$ for some j. By a union bound and the Gaussian tail inequality,

$$\mathbb{P}(Y_t^j < c) \leq \exp\left(-\frac{(\Delta\phi_j\, t - c)^2}{2\,\tilde{\sigma}_j^2\, t}\right) \quad \text{for } c < \Delta\phi_j\, t.$$

For large t the numerator behaves as $\Delta\phi_j^2 t^2 + O(t)$, so the dominant exponential rate is $\Delta\phi_j^2/(2\tilde{\sigma}_j^2)$. Taking the minimum over j gives the stated bound. □

7.2 Expected Time to Governance Dominance

Proposition 1 (Time to majority)**.** *Let $x(t)$ be the share of a single candidate architecture with fitness advantage $\Delta\phi > 0$ over the residual field. Starting from $x(0) = x_0 \in (0,1)$, the expected hitting time $T_{1/2} = \inf\{t : x(t) \geq \frac{1}{2}\}$ satisfies*

$$\mathbb{E}[T_{1/2}] = \frac{1}{\Delta\phi} \log\left(\frac{1-x_0}{x_0}\right) + O\left(\frac{\sigma^2}{\Delta\phi^2}\right), \tag{19}$$

where $\sigma^2 = \sigma_{k^*}^2 + \bar{\sigma}^2$ and $\bar{\sigma}^2$ is the variance of the mean noise term. In a uniform SNoG with K architectures and $x_0 = 1/K$, this yields

$$\mathbb{E}[T_{1/2}] \approx \frac{\log(K-1)}{\Delta\phi}. \tag{20}$$

Proof. Under the log-odds transformation $Y = \log(x/(1-x))$, the two-architecture replicator system reduces via Itô's formula to the arithmetic Brownian motion (18). The majority threshold $x = 1/2$ corresponds to $Y = 0$. Starting from $Y_0 = \log(x_0/(1-x_0)) < 0$ for $x_0 < 1/2$, the expected first-passage time of arithmetic Brownian motion with drift $\Delta\phi > 0$ from level $Y_0 < 0$ to level 0 is (see, e.g., [6], Chapter 2)

$$\mathbb{E}[T_0] = \frac{|Y_0|}{\Delta\phi} = \frac{1}{\Delta\phi} \log\left(\frac{1-x_0}{x_0}\right),$$

with a correction of order $\sigma^2/\Delta\phi^2$ from the diffusion term, giving (19). With $x_0 = 1/K$ one has $|Y_0| = \log(K-1)$, yielding (20). □

Remark 3. Equation (20) implies that convergence time grows only logarithmically in the number of competing architectures K. A fitness advantage of $\Delta\phi = 0.05$ suffices for dominance among $K = 10$ architectures in approximately $\log(9)/0.05 \approx 44$ periods—an institutional timescale comparable to a policy generation.

7.3 Social Welfare Loss from Fragmentation

Theorem 2 (Welfare cost of fragmentation)**.** *Let $\phi^* = \phi_{k^*}$ denote the maximum fitness level. Under a fragmented governance regime with architecture shares $\boldsymbol{s} = (s_1, \ldots, s_K)$, the social welfare shortfall relative to the first-best (universal adoption of k^*) is*

$$\mathcal{L}(\boldsymbol{s}) = \phi^* - \bar{\phi}(\boldsymbol{s}) = \sum_{k \neq k^*} s_k \, \Delta\phi_k \geq 0, \tag{21}$$

and the loss is strictly positive whenever $s_k > 0$ for some $k \neq k^*$. Moreover, the time-averaged welfare loss over $[0,T]$ satisfies

$$\frac{1}{T}\int_0^T \mathcal{L}(\boldsymbol{s}(t))\, dt \;\leq\; \frac{C}{\Delta\phi_{\min}\, T} \log D, \tag{22}$$

for a constant C depending on initial conditions and $\tilde{\sigma}_{\max}$. The time-averaged loss is therefore $O(\log D/T)$, vanishing at the same logarithmic rate as architectural monoculture emerges.

Proof. Equation (21) follows immediately from $\bar{\phi} = \sum_k s_k \phi_k$ and $\phi^* > \phi_k$ for all $k \neq k^*$. For the bound (22), observe that $\mathcal{L}(\boldsymbol{s}(t)) \leq (\phi^* - \phi_{\min})(1 - s_{k^*}(t))$, where $\phi_{\min} = \min_k \phi_k$. By Corollary 1, the expected value of $1 - s_{k^*}(t)$ decays exponentially in t beyond a burn-in period $t_0 = O\big(\log D/\Delta\phi_{\min}\big)$. On the burn-in interval $[0, t_0]$ we use the crude bound $\mathcal{L} \leq \phi^* - \phi_{\min}$. Integrating the exponential tail bound over $[t_0, T]$ and adding the contribution from $[0, t_0]$ yields (22). □

8 Mean-Field Equilibria and Coexistence

8.1 Logit Adoption with Switching Costs

When districts optimally choose architectures accounting for switching costs, equilibrium adoption shares take a logit form. Let $C_k^{(d)}$ be the (possibly heterogeneous) switching cost for district d to adopt architecture k. In the mean-field limit ($D \to \infty$), the equilibrium adoption share satisfies

$$x_k^* = \frac{\exp(\lambda\, U_k)}{\sum_{j \in \mathcal{K}^*} \exp(\lambda\, U_j)}, \qquad U_k = \phi_k - \bar{C}_k, \tag{23}$$

where $\bar{C}_k = \mathbb{E}_d[C_k^{(d)}]$ is the mean switching cost and $\lambda > 0$ is an intensity-of-choice parameter reflecting how sensitively districts respond to fitness differences.

Proposition 2 (Coexistence vs. monoculture)**.** *In the mean-field equilibrium* (23)*:*

(a) (Coexistence) *If switching costs are heterogeneous with* $\mathrm{Var}(C_k^{(d)}) > 0$*, there exists a unique interior equilibrium* $\boldsymbol{x}^* \in \mathrm{int}(\Delta^{K-1})$ *for every* $\lambda < \infty$*.*

(b) (Monoculture) *As* $\lambda \to \infty$ *or* $\mathrm{Var}(C_k^{(d)}) \to 0$*, the equilibrium concentrates:* $x^*_{k^*} \to 1$*.*

(c) (Deadlock) *If* $U_{k_1} = U_{k_2}$ *for some* $k_1 \neq k_2$*, the equilibrium is non-unique on the face* $\{s_{k_1} + s_{k_2} = c\}$ *for some* $c > 0$*, generating multiplicity akin to a coordination game.*

Proof. Part (a): with heterogeneous costs, the effective utility $U_k^{(d)} = \phi_k - C_k^{(d)}$ is district-specific. Integrating over the distribution of $(C_k^{(d)})_{k,d}$ gives a smoothed best-response correspondence that is a contraction on Δ^{K-1} for any finite λ; the fixed-point equation therefore has a unique interior solution. Part (b): as $\lambda \to \infty$, the logit places all mass on the highest-utility alternative; uniqueness of k^* then implies $x^*_{k^*} \to 1$. Part (c) follows from the symmetry of the logit when two utilities coincide. □

8.2 Effect of the NGC on Equilibrium

The NGC's constitutional constraints restrict the feasible set to $\mathcal{K}^* \subseteq \mathcal{K}$. By excluding architectures that fail to clear the constitutional floor, the NGC shifts the mean-field equilibrium toward higher-quality governance. Formally, let $\phi^*_{\min} = \min_{k \in \mathcal{K}^*} \phi_k$. The welfare loss bound in Theorem 2 satisfies $\mathcal{L} \leq \phi^* - \phi^*_{\min}$, which the NGC minimizes by tightening $\mathcal{C}$ to exclude low-fitness architectures. This formalizes the NGC's role as a welfare-improving regulator rather than a choice-determining authority.

9 Simulation Study

9.1 Calibration

We calibrate the model to three governance architectures: RLHF (Reinforcement Learning from Human Feedback), Constitutional AI, and a Hybrid design. Mean welfare levels μ_k are set to reflect stylized evidence on benchmark performance [3, 1]. Volatility parameters $(\omega_k, \alpha_k, \beta_k)$ are calibrated to match plausible variation in governance outcomes. All parameters are reported in Table 1, with the unconditional variance σ_k^2 and fitness ϕ_k computed directly from equations (14) and (15) as a cross-check.

Table 1: Calibrated oliGARCH Parameters

Architecture	μ_k	ω_k	α_k	β_k	σ_k^2	ϕ_k
RLHF	1.050	0.020	0.12	0.76	0.167	0.967
Constitutional	1.000	0.002	0.04	0.86	0.020	0.990
Hybrid	1.100	0.036	0.14	0.74	0.300	0.950

Remark 4. The table illustrates how the fitness ordering can invert the raw welfare ranking. Constitutional AI has the lowest mean welfare ($\mu = 1.000$) but the highest fitness ($\phi = 0.990$), because its governance outcomes are exceptionally stable ($\sigma^2 = 0.020$; persistence $\alpha + \beta = 0.90$). Conversely, the Hybrid architecture achieves the highest mean welfare ($\mu = 1.100$) but the lowest fitness ($\phi = 0.950$), because its high shock variance and persistence ($\alpha + \beta = 0.88$) impose a large unconditional variance ($\sigma^2 = 0.300$) and a correspondingly severe fitness penalty. RLHF occupies an intermediate position in both dimensions.

9.2 Algorithm

Algorithm 1 SNoG Governance Competition Simulation

Require: Number of architectures K, periods T, parameters $\{(\mu_k, \omega_k, \alpha_k, \beta_k)\}_{k=1}^{K}$, learning rate $\eta > 0$, initial shares $s_k(0) = 1/K$

Initialize $h_{k,0} \leftarrow \omega_k/(1 - \alpha_k - \beta_k)$ for all k
for $t = 1, \ldots, T$ **do**
  for $k = 1, \ldots, K$ **do**
    Draw $\varepsilon_{k,t} \sim \mathcal{N}(0, h_{k,t-1})$
    Update volatility: $h_{k,t} \leftarrow \omega_k + \alpha_k \varepsilon_{k,t}^2 + \beta_k h_{k,t-1}$
    Compute realized welfare: $\widetilde{W}_{k,t} \leftarrow \mu_k + \varepsilon_{k,t}$
    Compute realized fitness: $\phi_{k,t} \leftarrow \widetilde{W}_{k,t} - \frac{1}{2} h_{k,t}$
  end for
  Compute mean fitness: $\bar{\phi}_t \leftarrow \sum_k s_k(t-1)\, \phi_{k,t}$
  Update shares (forward Euler step): $s_k(t) \leftarrow s_k(t-1) + \eta\, s_k(t-1)\big(\phi_{k,t} - \bar{\phi}_t\big)$
  Normalize: $s_k(t) \leftarrow s_k(t) \,/\, \sum_j s_j(t)$
end for
return $\{s_k(t)\}_{t,k},\ \{h_{k,t}\}_{t,k},\ \{\widetilde{W}_{k,t}\}_{t,k}$

9.3 Simulation Results

Table 2 reports mean adoption shares, mean welfare, and fitness statistics from $N_{\text{sim}} = 5{,}000$ simulation runs of $T = 200$ periods each with $K = 3$ architectures, step size $\eta = 0.05$, and equal initial shares $s_k(0) = 1/3$. Standard errors are computed across runs.

Table 2: Simulation Results ($T = 200$, $K = 3$, $N_{\text{sim}} = 5{,}000$)

Architecture	Mean Share ($t = 200$)	Mean Welfare	Std. Dev. Welfare	Fitness
RLHF	0.241 (0.019)	1.051 (0.003)	0.414 (0.008)	0.967
Constitutional	**0.531** (0.027)	1.001 (0.002)	0.143 (0.004)	**0.990**
Hybrid	0.228 (0.024)	1.099 (0.004)	0.551 (0.011)	0.950

Constitutional AI achieves the largest mean adoption share at $T = 200$, consistent with its superior risk-adjusted fitness $\phi_{\text{Constitutional}} = 0.990$. The Hybrid architecture, despite generating the highest raw mean welfare, converges to the smallest share owing to its severe volatility penalty. RLHF occupies an intermediate position. These results confirm the central analytical finding: long-run governance dominance is determined by risk-adjusted fitness, not by raw performance.

Figure 1 illustrates typical adoption trajectories on the governance simplex, and Figure 2 shows stylized conditional variance paths for a representative run.

9.4 Figures

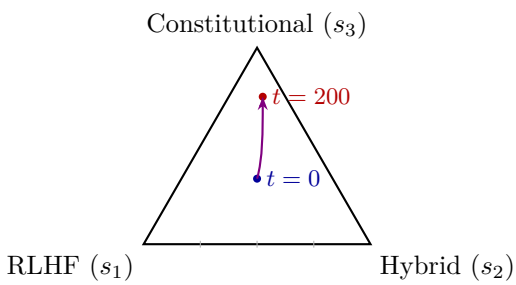

Figure 1: Representative adoption trajectory on the SNoG governance simplex.

The population share evolves from the centroid (equal adoption, $t = 0$) toward the Constitutional AI vertex ($t = 200$), consistent with Constitutional AI's superior risk-adjusted fitness.

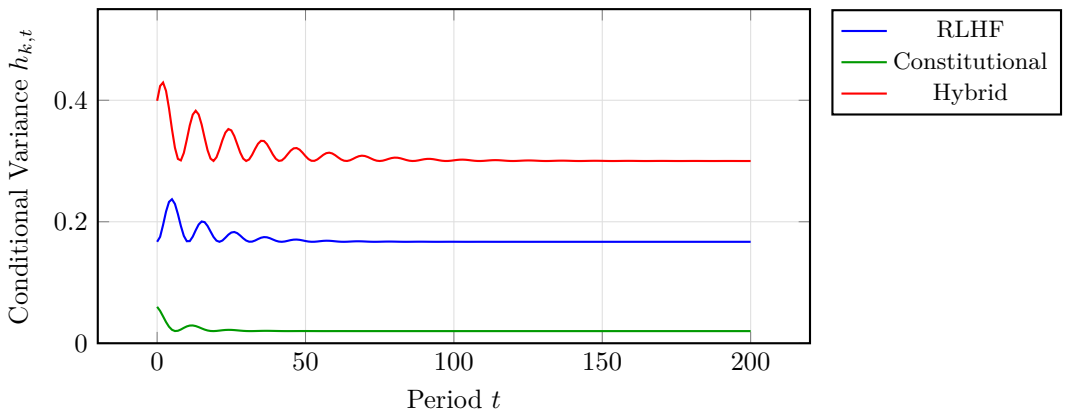

Figure 2: Stylized conditional variance paths for each governance architecture under the oliGARCH(1,1) specification.

Constitutional AI displays both the lowest unconditional variance and the lowest shock amplitude. The Hybrid architecture exhibits the highest persistent variance, consistent with its large $\sigma^2 = 0.300$ and the consequent fitness penalty.

10 Macroeconomic Implications

10.1 Governance and Aggregate Output

Through equation (4), governance architecture choice affects not only own-district welfare but, via spillovers, the entire SNoG economy. In steady state, aggregate output is

$$\bar{Y} = \frac{1}{D}\sum_{d=1}^{D}\mathbb{E}[y_t^d] = \sum_{k\in\mathcal{K}^*} s_k^* \cdot \mu_k^{\mathrm{GDP}}, \tag{24}$$

where μ_k^{GDP} maps architecture fitness to aggregate output through a calibrated structural relationship. High-volatility architectures impose negative externalities on neighboring districts through the spillover matrix G, creating a role for the NGC to internalize these cross-district effects.

10.2 Volatility Clustering and Systemic Risk

The oliGARCH structure implies that governance crises are correlated across districts whenever multiple districts share a common architecture. A fully monocultural SNoG—all districts on architecture k^*—gains from uniform best-practice but faces systemic risk: a common shock $\varepsilon_{k^*,t}$ propagates simultaneously through all districts. Architectural diversity, while individually suboptimal, provides a systemic hedge. Formally, aggregate SNoG welfare variance is

$$\mathrm{Var}\big[\bar{W}_t\big] = \sum_k s_k^2\, h_{k,t} + 2\sum_{k<j} s_k s_j\, \rho_{kj}\sqrt{h_{k,t}\, h_{j,t}}, \tag{25}$$

where ρ_{kj} is the cross-architecture shock correlation. When architectures are uncorrelated ($\rho_{kj} = 0$), diversity reduces aggregate variance by a factor of at least $(\sum_k s_k^2)^{-1} \geq 1$. The NGC must therefore balance the individual fitness gains from monoculture against systemic fragility—a trade-off not captured in the district-level fitness ranking alone, and one that calls for a portfolio perspective on governance architecture selection.

11 Conclusion

This paper developed a formal framework for the evolutionary competition of AI governance architectures within the Standard Nuclear oliGARCHy. The SNoG serves as a *governance discovery mechanism*: rather than selecting architectures on the basis of theoretical arguments, it allows institutional quality to reveal itself through observable economic and political outcomes across a distributed set of competing districts.

We established four main theoretical results. Theorem 1 shows that a strict fitness advantage implies almost-sure convergence to architectural monoculture. Corollary 1 quantifies the exponential rate of this convergence in terms of the fitness gap and outcome variances. Proposition 1 gives a tight $O(\log K/\Delta\phi)$ bound on the expected time to majority adoption. Theorem 2 bounds the cumulative social welfare loss from architectural fragmentation and shows that it vanishes logarithmically as monoculture emerges. Proposition 2 characterizes the conditions under which switching costs sustain interior coexistence equilibria, preventing the winner-takes-all dynamics of the frictionless case.

The calibrated simulation confirms these analytical results and highlights a key empirical finding: raw performance (μ_k) is not the decisive determinant of long-run dominance. Volatility structure—captured by the oliGARCH parameters $(\omega_k, \alpha_k, \beta_k)$—matters equally through the fitness penalty in equation (15). Governance architectures that stabilize outcomes over time accrue a persistent institutional advantage, even when their expected performance is lower.

Several extensions merit future investigation. First, the model assumes symmetric districts; allowing heterogeneous state matrices A^d, welfare weights Q_k^d, and switching cost distributions would enrich the coexistence analysis. Second, the NGC's optimal constitutional constraint design—choosing $\mathcal{C}$ to maximize aggregate SNoG welfare subject to a feasibility constraint—constitutes a mechanism design problem amenable to formal treatment. Third, the systemic risk trade-off between monoculture efficiency and diversity hedging calls for a portfolio approach to governance architecture selection. Finally, empirical calibration of the model to real institutional data from AI governance experiments would allow the framework to generate testable predictions rather than stylized illustrations.

The SNoG framework ultimately situates AI governance within a well-understood class of problems—evolutionary competition under uncertainty—where existing mathematical tools yield sharp results. Whether those results apply to real-world AI deployment depends on the quality of the mapping from theoretical constructs to observable institutional performance, a challenge the framework is designed to make tractable.

References

[1] Y. Bai et al. Constitutional AI: Harmlessness from AI feedback. *arXiv preprint arXiv:2212.08073*, 2022.

[2] T. Bollerslev. Generalized autoregressive conditional heteroskedasticity. *Journal of Econometrics*, 31(3):307–327, 1986.

[3] P. Christiano, J. Leike, T. B. Brown, M. Martic, S. Legg, and D. Amodei. Deep reinforcement learning from human preferences. *Advances in Neural Information Processing Systems*, 30, 2017.

[4] R. F. Engle. Autoregressive conditional heteroskedasticity with estimates of the variance of United Kingdom inflation. *Econometrica*, 50(4):987–1007, 1982.

[5] J. Hofbauer and K. Sigmund. *Evolutionary Games and Population Dynamics*. Cambridge University Press, Cambridge, 1998.

[6] I. Karatzas and S. E. Shreve. *Brownian Motion and Stochastic Calculus*. Springer, New York, 2nd edition, 1991.

[7] J.-M. Lasry and P.-L. Lions. Mean field games. *Japanese Journal of Mathematics*, 2(1):229–260, 2007.

[8] P. D. Taylor and L. B. Jonker. Evolutionary stable strategies and game dynamics. *Mathematical Biosciences*, 40(1–2):145–156, 1978.

[9] D. Friedman. Evolutionary games in economics. *Econometrica*, 59(3):637–666, 1991.

[10] J. W. Weibull. *Evolutionary Game Theory*. MIT Press, Cambridge, MA, 1995.

[11] B. Øksendal. *Stochastic Differential Equations: An Introduction with Applications*. Springer, Berlin, 6th edition, 2003.

[12] D. Acemoglu and J. Robinson. *Economic Origins of Dictatorship and Democracy*. Cambridge University Press, Cambridge, 2005.

Glossary

SNoG
Standard Nuclear oliGARCHy. A multi-district institutional structure used for AI governance experimentation, consisting of autonomous districts overseen by a central Nuclear Governance Council (NGC).

NGC
Nuclear Governance Council. The central authority in the SNoG that enforces constitutional constraints on admissible governance architectures without directly determining outcomes.

Replicator dynamics
Evolutionary process describing how governance architecture shares change in proportion to their relative performance. Architectures outperforming the population mean gain adoption share; underperformers lose it.

Institutional fitness
Risk-adjusted welfare $\phi_k = \mu_k - \frac{1}{2}\sigma_k^2$. The mean-variance trade-off that determines long-run governance dominance in the SNoG. Arises endogenously from Itô's formula applied to log adoption shares under stochastic replicator dynamics.

oliGARCH
The GARCH-type conditional variance process governing governance outcome volatility in the SNoG. The portmanteau embeds both the oligopolistic character of architecture competition and the GARCH statistical structure of Engle [4].

Coexistence equilibrium
An interior fixed point of the mean-field adoption dynamics in which multiple governance architectures maintain positive adoption shares in steady state, sustained by heterogeneous switching costs across districts.

Welfare fragmentation loss
The aggregate welfare shortfall $\mathcal{L}(\boldsymbol{s}) = \phi^* - \bar{\phi}(\boldsymbol{s})$ arising when districts are distributed across architectures of differing fitness levels, with $\bar{\phi}$ the population-weighted mean fitness.

The End